

# Todo App V2 (Individual Version)

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## Application Flow

This app is a browser-based todo list that stores data in LocalStorage and updates the UI after every change.

### 1. Page load

- The browser loads index.html.
- script.js runs after HTML parsing because the script tag uses defer.
- DOMContentLoaded fires.
- logEvent logs the DOMContentLoaded event to the on-screen monitor.
- loadTodosFromStorage reads LocalStorage key todos.
- loadTodosFromStorage also prints the storage snapshot to the on-screen monitor.
- renderTodos displays either the empty state or todo items.

### 2. User submits a new todo

- submit event is captured on the form.
- handleFormSubmit prevents default form reload.
- Input is trimmed and validated.
- If invalid, an error message is shown.
- If valid, addToDo creates a todo object and pushes it into the todos array.
- updateUI is called.
- updateUI calls saveTodosToStorage, then renderTodos.
- saveTodosToStorage logs LocalStorage before save and after save.

### 3. User toggles a todo checkbox

- change event is handled through event delegation on the todo list.
- handleTodoToggle finds the target todo id.
- toggleTodo updates completed for the matching item.
- updateUI saves and re-renders.
- Storage snapshots are printed before and after save.

### 4. User deletes a todo

- click event is handled through event delegation on the todo list.
- handleTodoListClick checks for delete button clicks.
- deleteTodo removes the matching item from the array.
- updateUI saves and re-renders.
- Storage snapshots are printed before and after save.

### 5. Event monitor logging

- logEvent prints event details including:
  - event name
  - phase (capturing, at\_target, bubbling)
  - currentTarget

- target
- log appends lines into the monitor area on screen.
- Clear Log button resets step count and clears monitor output.

## Data Model

Each todo is stored as:

- id: string (timestamp)
- text: string
- completed: boolean
- createdAt: ISO datetime string

## LocalStorage

- Key: todos
- Value: JSON stringified array of todo objects

## Sequence Diagram

```
sequenceDiagram
    participant U as User
    participant B as Browser
    participant JS as script.js
    participant LS as LocalStorage
    participant UI as DOM/UI

    B->>JS: DOMContentLoaded
    JS->>UI: logEvent("DOMContentLoaded")
    JS->>LS: getItem("todos")
    LS-->>JS: stored todos or null
    JS->>UI: logLocalStorageState("load from storage")
    JS->>UI: renderTodos()

    U->>UI: Submit form
    UI->>JS: submit event
    JS->>JS: handleFormSubmit()
    JS->>JS: addTodo()
    JS->>JS: updateUI()
    JS->>LS: getItem("todos")
    JS->>UI: logLocalStorageState("before save")
    JS->>LS: setItem("todos", JSON)
    JS->>LS: getItem("todos")
    JS->>UI: logLocalStorageState("after save")
    JS->>UI: renderTodos()

    U->>UI: Toggle checkbox
    UI->>JS: change event
    JS->>JS: handleTodoToggle()
    JS->>JS: toggleTodo()
    JS->>JS: updateUI()
```

```
JS->>LS: setItem("todos", JSON)
JS->>UI: renderTodos()

U->>UI: Click Delete
UI->>JS: click event
JS->>JS: handleTodoListClick()
JS->>JS: deleteTodo()
JS->>JS: updateUI()
JS->>LS: setItem("todos", JSON)
JS->>UI: renderTodos()
```